The Gough Map: Britain's Oldest Road Map, Measured

Shape and distance on a map of about 1360:
157 way stations located on the parchment and 169 written mileages checked against the roads

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Abstract

The Gough Map (Bodleian Library, MS. Gough Gen. Top. 16), drawn about 1360 with east at the top, is the oldest surviving map of Britain to show a road network: red lines join way stations and most carry a distance in Roman numerals. I combine the GIS database of Oksanen and Brookes, which gives the true positions of its 179 way stations, its 188 red lines with their figures and the real routes they most likely stand for, with the positions of 157 way stations (88%) located by eye on a high-resolution photograph of the map. A similarity fit turns the map 95° from north-up; at its average scale the median way station is misplaced by 17.6 km, and an affine fit adds little (stretch 1.11). Wales is drawn at 1.22 times the average scale with a median error of 33.2 km; Scotland, checked with a few control points, is compressed. One written mile corresponds to a median 2.08 km of road (1.29 statute miles) and 1.89 km in a straight line; 72% of 169 figures are within 25% of the road length at that rate. The written figures are more accurate than the drawn lengths of the same lines (median error 13% against 31%), and summed along the network they give London to York as 420 km against 424 km of road. Of the 4,505 km of road under the red lines, 35.4% is Roman; the figures are equally good on Roman and later roads.

Keywords: Gough Map; medieval cartography; road networks; itineraries; map distortion; historical GIS

1 Introduction

The Gough Map is a map of Great Britain on two pieces of parchment, about 115 by 56 cm, in the Bodleian Library, Oxford. It is named after the antiquary Richard Gough, who acquired it in 1774, and is usually dated to about 1360, with later revisions (Parsons, 1958; Delano-Smith et al., 2017). It is oriented with east at the top. Its most celebrated feature is a network of thin red lines joining towns, most of them with a distance written beside them in Roman numerals: the earliest surviving road map of Britain (Lilley et al., 2009; Delano-Smith et al., 2017). How the network was compiled, and how it relates to real roads, has long been debated; Oksanen and Brookes (2024) recently published a GIS database that identifies, for each red line, the most likely real route on the ground, and dates its parts from Roman roads to the road books of the seventeenth and eighteenth centuries.

This paper uses that database to ask two questions that need both the map and the ground: how wrong is the map's shape, region by region, and how right are its written distances? The first requires the positions of the way stations on the map itself, which the database does not give; I located them by eye. The study accompanies an interactive comparison of the map and an ink map of the same roads (https://maps.tristanhotham.com/gough-map/).

2 Data

The road database. Oksanen and Brookes (2024): 179 way stations (settlements joined by a red line) in British National Grid, with modern name, historic county and legibility of the name on the map; 188 red lines between them with the written figure (19 absent or faded); and 455 route segments, about 4,542 km in all, each with a period code, a category (Roman road, a road on Cary's maps, modern road), Margary numbers for Roman roads (Margary, 1973), and the evidence used. The period codes are numeric and I could not reach the published key; from the categories and comments I read them as: 1 Roman road; 2 evidence before Domesday (1086), such as Anglo-Saxon charter bounds; 3 dated medieval evidence after Domesday, mostly bridges; 4 other medieval evidence, such as the routes of Gerald of Wales or the 1334 lay subsidy; 5 known only from the early modern road books (Ogilby, 1675); 6 river; 0 undated or modern road. Results by period depend on that reading.

The map. A photograph of the whole map, 12,224 by 6,021 pixels, from Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Gough_Kaart_(hoge_resolutie).jpg), a faithful reproduction of a public domain work. **Outlines.** Natural Earth (Natural Earth, n.d.).

3 Methods

3.1 Locating the way stations

No published table gives where each way station is drawn. I located them on the photograph at full resolution, on gridded crops. I started from unmistakable towns (London, York, Exeter, St David's), fitted an affine and then a thin-plate spline from the National Grid to the image, and used it to predict where each remaining town should be drawn. Each was then identified on the crop: by its label where legible (33, "high"), by the right town drawing at the end of the right red line with the right numeral (56, "medium"), or with real doubt, usually among faded drawings or several candidates (68, "low"). 22 could not be placed, 6 of them in Cornwall, where the parchment is worn. Because the map has no red lines in Scotland, I added 6 control points read the same way: Orkney, the Isle of May, the Bass Rock, Perth and the two ends of Hadrian's Wall (Table 2). As a consistency check, a smooth fit through all other points predicts each point to within a median 8.1 km.

3.2 Shape

Positions in National Grid kilometres were fitted to image pixels (with the image y axis turned up) by least squares, first with a similarity (one scale, one rotation, a shift), then with a general affine transformation (Tobler, 1994). A way station's misplacement is the distance between its true position and its map position taken back to the ground through the inverse of the overall similarity: how far the map puts it from where a correctly shaped map at the same scale and orientation would. Each region of at least six located places (groups of historic counties) has its own similarity fit, reported as its scale relative to the whole map and its rotation relative to the map's mean rotation, and an affine fit whose ratio of singular values measures stretch. For the figures, a thin-plate spline (Bookstein, 1989) with light smoothing carries the National Grid and the modern coastline onto the image, and back again at the overall similarity (the "Gough shape" of England and Wales). The spline is unconstrained far from located points and is not used for Scotland.



Figure 1: The Gough Map with the way stations located (black rings), the control points (blue) and the National Grid at 25 km carried onto the image by a thin-plate spline. East is at the top.

3.3 Distances

The route segments were joined into a network at the vertex level (vertices within 30 m merged; loose ends joined to the nearest vertex of another segment within 400 m). Each way station was put on the nearest vertex (median offset under 0.1 km, at most 1.7 km, added to the length), and each red line's road distance is the shortest path between its two way stations over segment lengths as given in the database. Straight-line distances are in the National Grid. For each of the 169 red lines with a legible figure, kilometres per written mile is road (or straight) distance divided by the figure; a figure "agrees within 25%" when the figure times the median rate is within a factor of 1.25 of the road distance. The drawn length of a red line is the image distance between its two located stations divided by the overall similarity scale. Journeys are shortest paths over the red lines weighted by the written figure, with faded figures filled from the road length at the median rate.

4 Results

4.1 The two maps

Figure 1 shows the located way stations on the photograph with the 25 km National Grid bent to fit. The grid is remarkably regular over England, tightening towards the north and opening out over Wales.

4.2 Shape

The similarity fit turns the map 95° anticlockwise from north-up (96° with high and medium confidence points only): east at the top, with a further lean of about 4.6°. At the overall scale,

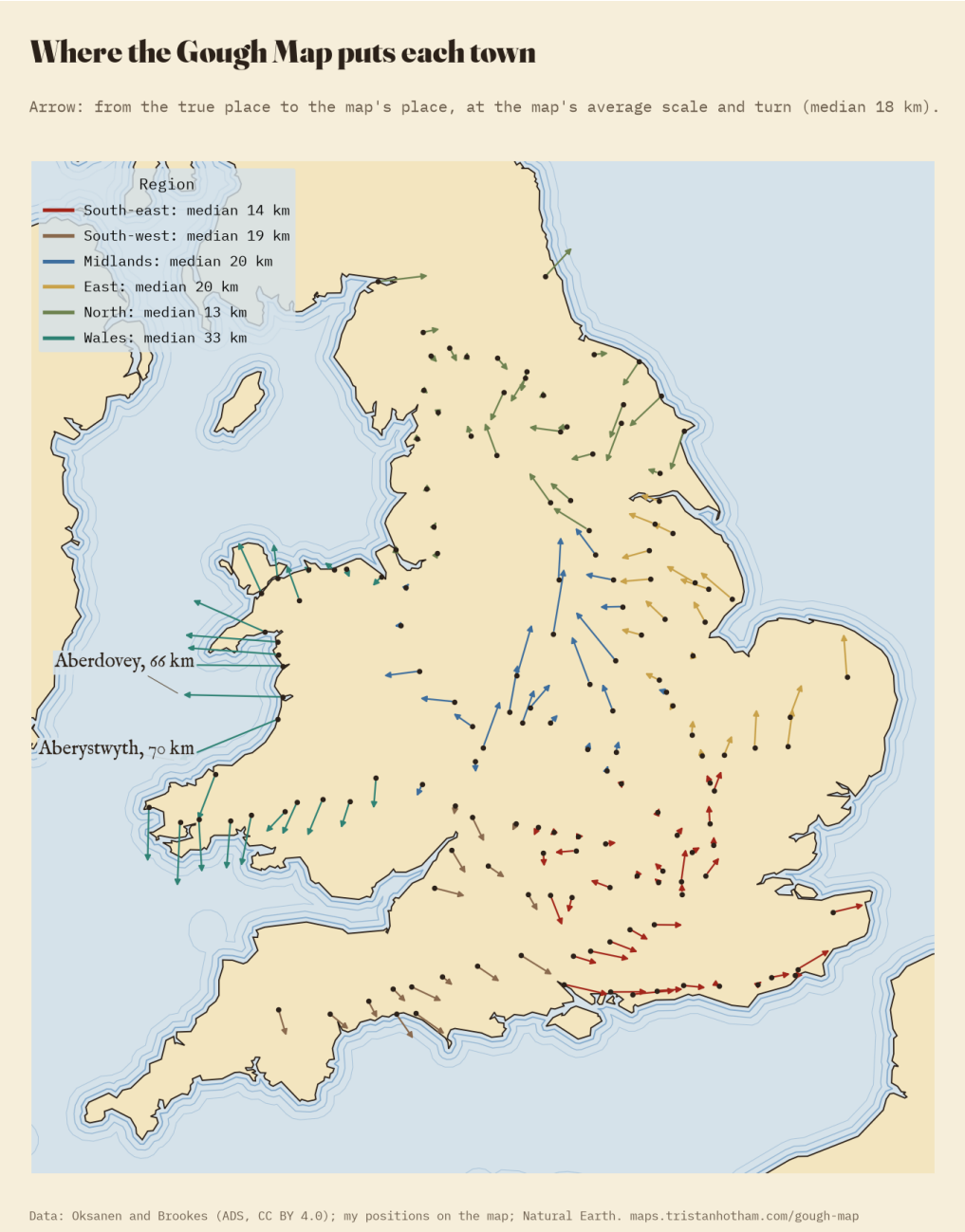


Figure 2: Misplacement of each located place: arrow from the true position to the map position at the overall scale and rotation, coloured by region.

the median way station is misplaced by 17.6 km and 76% are within 25 km; the root mean square error is 31.3 km. An affine fit barely improves on this (stretch 1.11; 1.07 for the surer points), so the map has no strong overall squeeze in one direction. Errors vary by region (Figure 2, Table 1). The north of England (12.6 km) and the south-east (13.9 km) are best; the Midlands, East and South-west lie near 20 km. Wales is drawn at 1.22 times the overall scale and its way stations are misplaced by a median 33.2 km; the five worst-placed way stations are all on the coast of Cardigan Bay, led by Aberystwyth (70 km). The north of England is drawn small (0.90).

North of the Wall the map is much less reliable. The Wall's ends are within about 31 km of their places. Orkney is 450 km from the middle of the Wall but drawn at 356 km (0.79), and is misplaced by 246 km; Perth is drawn at 0.67 of its distance, while the Isle of May and the Bass Rock, at the mouth of the Forth, are pushed out (1.24 and 1.10). With four points these are indications only.

Table 1: Regions (groups of historic counties): located places; median misplacement at the overall similarity (km); RMS residual of the region's own similarity fit (km); its scale relative to the whole map; its rotation relative to the whole map (degrees, anticlockwise positive); and the stretch of its affine fit.

Region	Places	Misplaced	RMS	Scale	Turn	Stretch
South-east	39	13.9	10.3	1.04	+7.6	1.09
South-west	17	19.1	7.4	1.01	+3.0	1.11
Midlands	25	20.3	17.8	1.02	+4.9	1.07
East	21	19.6	7.4	1.01	+10.7	1.13
North	33	12.6	15.7	0.90	-6.7	1.10
Wales	24	33.2	22.4	1.22	+5.2	1.23

Figure 3 shows England and Wales in the map's shape.

4.3 The written miles

One written mile corresponds to a median 2.08 km of road (interquartile range 1.85 to 2.37) and 1.89 km in a straight line, or 1.29 and 1.17 statute miles; the roads are a median 1.10 times the straight line. At 2.08 km a mile, 72% of the 169 figures agree with the road within 25% and 89% within 50%; the rank correlation between figure and road length is 0.74 (Figure 4). The rate is lower for long lines (1.93 km for figures above 12) than for short ones (2.21 km for 8 or less), and ranges by region from 1.86 (South-west) to 2.34 (Midlands). The largest misfits (Rye to Winchelsea, 8 miles for 4.7 km; Barmouth to Llanenddwyn, 12 for 7.7) look like copying errors or misattributed figures.

4.4 Drawing against numbers

For 146 lines with both ends located and a figure, the written figure is closer to the road length than the drawn length on 66% of lines; median absolute errors are 13% and 31%, and 71% of figures but only 47% of drawn lengths are within 25% (Figure 5). Rank correlations with the road are 0.72 and 0.47.

4.5 Journeys

The red lines form 4 connected groups; the largest joins 154 way stations. Summing written figures along the shortest routes from London gives York at 202 miles (420 km against 424 km of road), Bristol at 202 against 203, Carlisle 485 against 493, St David's 445 against 432 and Exeter 343 against 292, the largest discrepancy (Figure 6).

4.6 Roads by period

The red lines use 436 route segments, 4,505 km. By period code, 35.4% is Roman road, 13.1% has evidence before 1086, 6.7% dated medieval evidence, 9.1% other medieval evidence and 32.7% is known only from the early modern road books (Figure 7). 71 red lines (38%) run mostly on Roman road. The rate per written mile is 2.10 km on mainly Roman lines and 2.10 km on lines known only from the road books, with 71% and 68% within 25%.

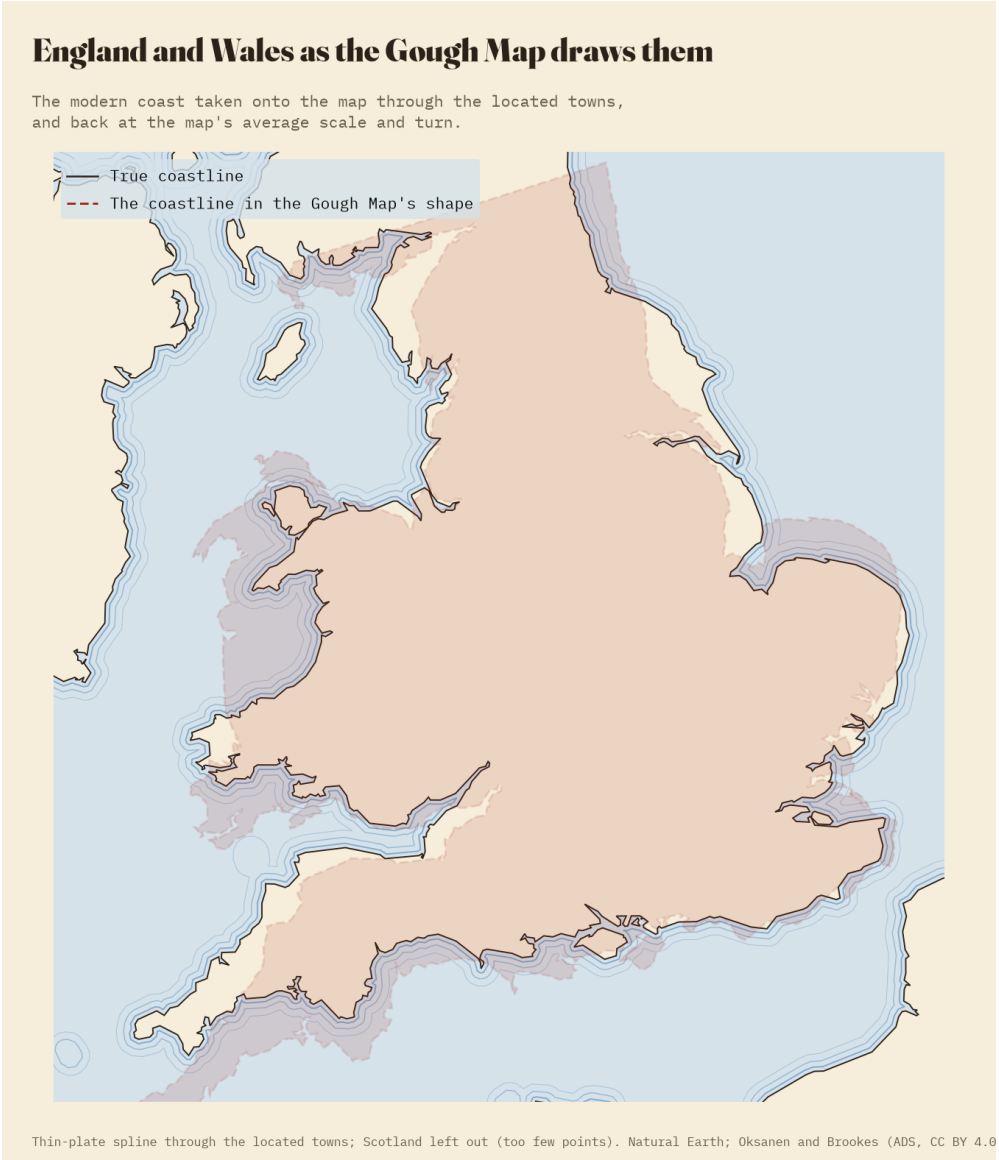


Figure 3: The modern coastline of England and Wales carried onto the map by the thin-plate spline and back to the ground at the overall similarity: the island as the Gough Map draws it.

5 Limitations

The positions on the map are mine, read by eye; 68 are uncertain and 22 are missing, and they favour legible places. Leaving the uncertain ones out moves the rotation by under two degrees and changes no conclusion, but individual misplacements, especially in Wales and the far north of England, should be read with that in mind. The road distances depend on Oksanen and Brookes' identification of the routes, which is itself an argument, and on my reading of the period codes. The similarity and affine fits summarise a map that was never drawn to one projection; the thin-plate spline is only a visual aid. The Scottish control points are few, and some are guesses. The statute mile is used only as a yardstick: the map's mile was a customary unit.



Figure 4: Written figure against road length for each red line with a legible figure, coloured by the period code covering most of the road. The band is agreement within 25% at the median rate.

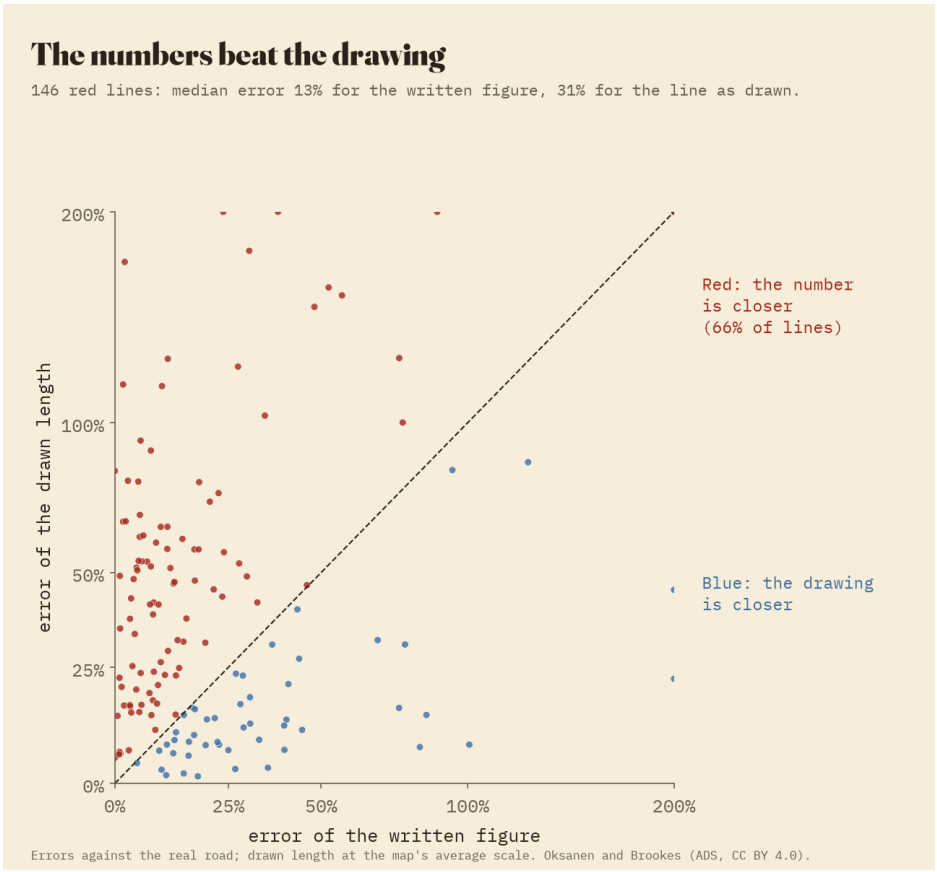


Figure 5: Absolute error of the written figure (at the median rate) against absolute error of the drawn length, per red line.

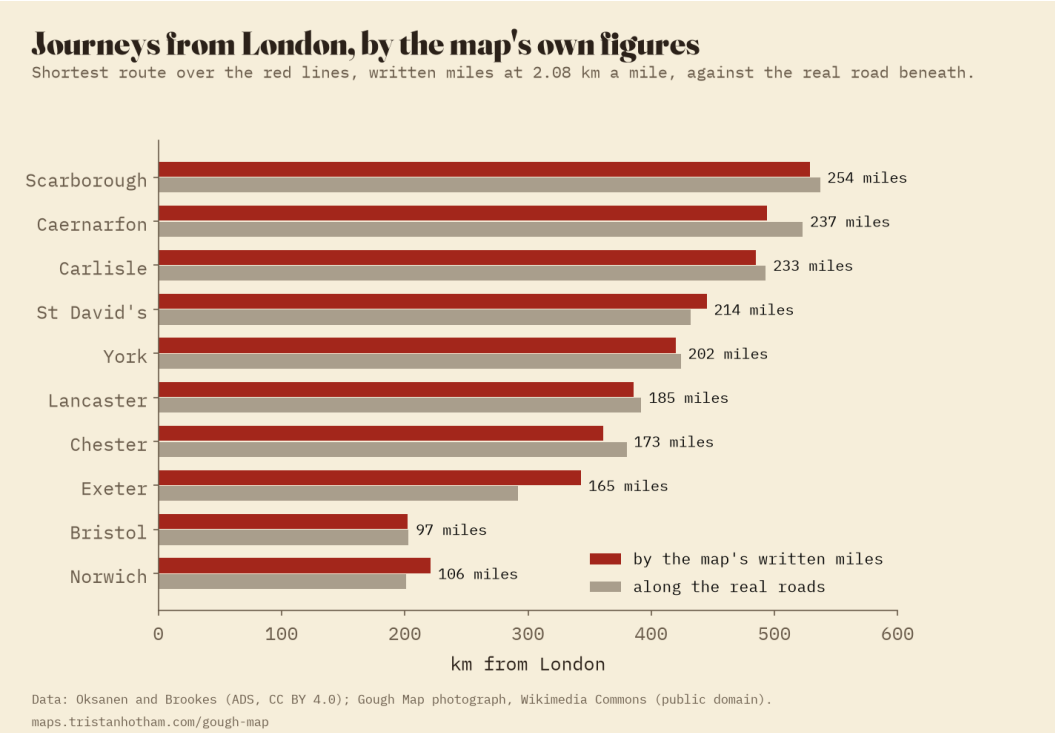


Figure 6: Journeys from London: written figures summed along the shortest route over the red lines and converted at the median rate, against the road length along the same lines.

6 Conclusion

The Gough Map places the towns of England within a couple of tens of kilometres of where an accurate map at its scale would, and its errors are regional: Wales enlarged and spread, the north slightly shrunk, Scotland squeezed. Its written distances are more reliable than its drawing: in a medieval mile of about 2.08 km of road, most are within a quarter of the truth, and over long journeys the errors cancel. Whoever compiled the red lines worked from distances along roads, many of them Roman, many first recorded centuries later.

Data and code availability

Data, code, the interactive page and the positions on the map are at https://github.com/Tristan-Hotham/gough-map and https://maps.tristanhotham.com/gough-map/, under the MIT licence (code) and CC BY 4.0 (text, figures and derived data). The road database is CC BY 4.0 (Oksanen and Brookes, 2024); the photograph is public domain.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

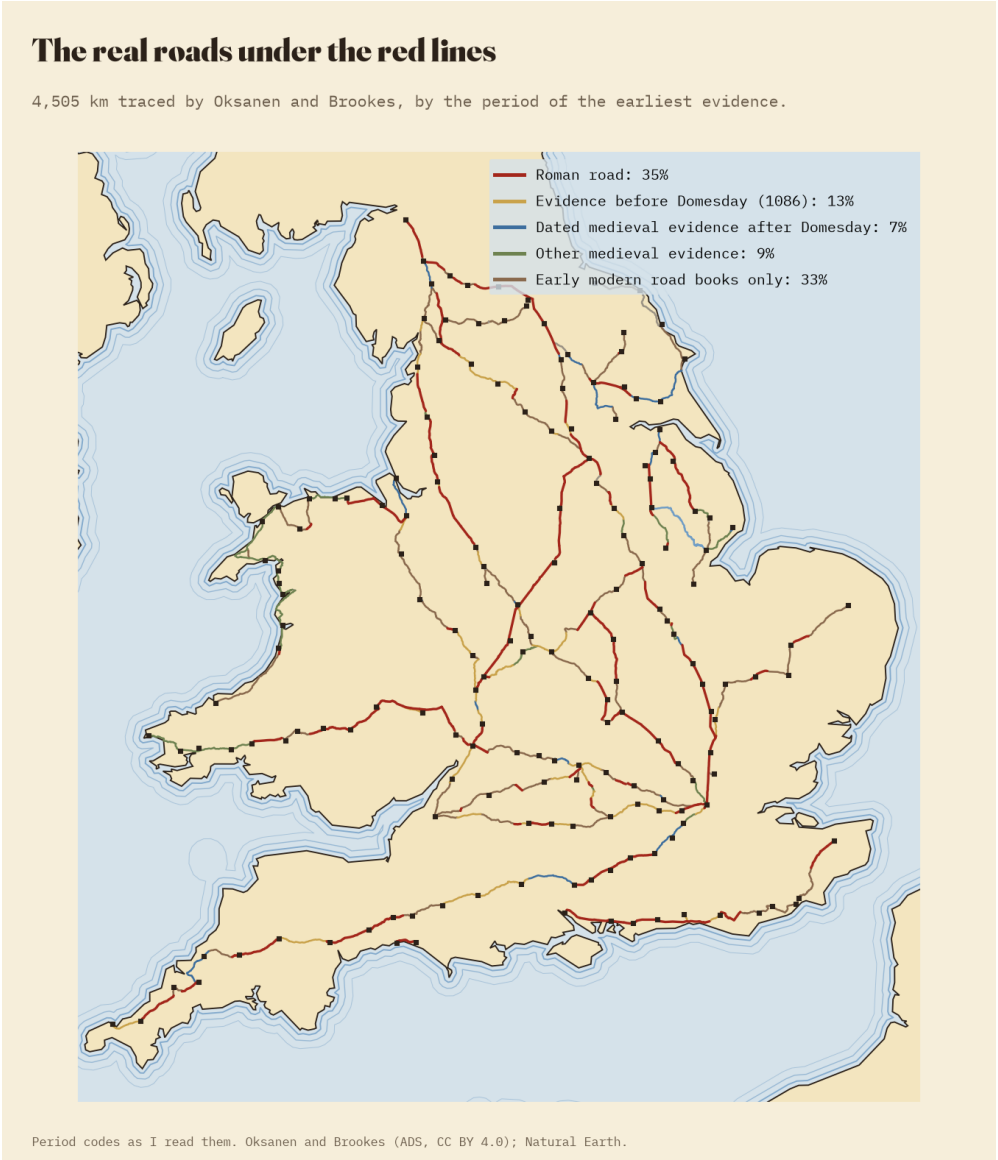


Figure 7: The route segments under the red lines by period code (my reading of the codes), with the way stations.

References

Bookstein, F. L. (1989) Principal warps: thin-plate splines and the decomposition of deformations. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11(6), 567-585.

Delano-Smith, C. et al. (2017) New light on the medieval Gough Map of Britain. *Imago Mundi* 69(1), 1-36.

Lilley, K. D., Lloyd, C. D. and Campbell, B. M. S. (2009) Mapping the realm: a new look at the Gough map of Britain (c. 1360). *Imago Mundi* 61(1), 1-28.

Margary, I. D. (1973) *Roman Roads in Britain*. 3rd edn. London: John Baker.

Natural Earth. Free vector and raster map data at 1:10m, 1:50m and 1:110m scales. Website, https://www.naturalearthdata.com/ (accessed 2026). Public domain.

Ogilby, J. (1675) *Britannia*. London.

Oksanen, E. and Brookes, S. (2024) *The Routes and Roads of the Gough Map: GIS Database* [data-set]. York: Archaeology Data Service [distributor]. https://doi.org/10.5284/1124312.

Parsons, E. J. S. (1958) *The Map of Great Britain circa A.D. 1360 known as the Gough Map: An Introduction to the Facsimile*, with "The roads of the Gough Map" by Sir Frank Stenton. Oxford: Bodleian Library and Royal Geographical Society.

Tobler, W. R. (1994) Bidimensional regression. *Geographical Analysis* 26(3), 187-212.

Table 2: The 163 places located on the photograph of the Gough Map (12,224 by 6,021 pixels; x to the right, y down), with confidence and misplacement (km) at the overall similarity, by region and name.

Place	County	x	y	Read	km
Barton-upon-Humber	Lincolnshire	6,600	2,320	med	14
Boston	Lincolnshire	7,230	1,925	med	17
Bury St Edmunds	Suffolk	7,860	1,230	high	26
Caistor	Lincolnshire	6,770	2,230	med	16
Cambridge	Cambridgeshire	8,060	1,595	high	16
Caxton	Cambridgeshire	8,155	1,800	high	7
Glanford Brigg	Lincolnshire	6,720	2,390	med	20
Grantham	Lincolnshire	7,460	2,375	med	15
Horncastle	Lincolnshire	7,030	2,120	med	23
Huntingdon	Huntingdonshire	7,980	1,855	high	11
Lincoln	Lincolnshire	7,165	2,410	med	22
Newmarket	Cambridgeshire	7,880	1,460	high	27
Norwich	Norfolk	7,360	950	high	30
Ogerston	Huntingdonshire	7,970	2,020	med	6
Old Bolingbroke	Lincolnshire	7,110	2,030	low	22
Sleaford	Lincolnshire	7,254	2,235	low	20
Spalding	Lincolnshire	7,550	1,905	low	5
Spital-in-the-Street	Lincolnshire	7,020	2,410	med	20
Stamford	Lincolnshire	7,720	2,200	med	12
Thetford	Norfolk	7,700	1,180	med	24
Wainfleet	Lincolnshire	7,050	1,900	med	28
Birmingham	Warwickshire	7,720	2,990	low	33
Blyth	Nottinghamshire	6,830	2,710	low	24
Bridgnorth	Shropshire	8,000	3,610	low	24
Chester	Cheshire	7,280	3,720	high	3
Chesterfield	Derbyshire	6,890	2,785	low	30
Coleshill	Warwickshire	7,830	2,765	low	22
Coventry	Warwickshire	8,020	2,700	med	9
Daventry	Northamptonshire	8,170	2,490	med	7
Derby	Derbyshire	7,090	2,746	low	44
Droitwich	Worcestershire	7,960	3,080	low	34
Hereford	Herefordshire	8,640	3,570	high	11
Kidderminster	Worcestershire	8,070	3,390	low	17
Leicester	Leicestershire	7,515	2,672	med	34
Lichfield	Staffordshire	7,460	2,920	low	36
Market Harborough	Leicestershire	7,800	2,455	low	20
Melton Mowbray	Leicestershire	7,362	2,665	low	42
Newark-on-Trent	Nottinghamshire	7,335	2,520	high	16
Northampton	Northamptonshire	8,150	2,310	high	9
Overton	Cheshire	7,560	3,810	med	6
Shrewsbury	Shropshire	7,880	3,850	med	25
Solihull	Warwickshire	7,870	2,870	low	24
Towcester	Northamptonshire	8,330	2,370	low	3
Tuxford	Nottinghamshire	7,130	2,630	low	20
Wansford	Northamptonshire	7,820	2,110	high	7
Worcester	Worcestershire	8,460	3,200	high	9
Appleby-in-Westmorland	Westmorland	5,870	3,520	low	12
Beverley	Yorkshire, East Riding	6,430	2,290	med	10
Boroughbridge	Yorkshire, West Riding	6,230	3,060	low	22
Bowes	Yorkshire, North Riding	5,895	3,190	low	12
Bridlington	Yorkshire, East Riding	6,450	2,130	low	29
Brough	Westmorland	5,770	3,470	low	5
Castle Bolton	Yorkshire, North Riding	6,230	3,295	low	22
Doncaster	Yorkshire, West Riding	6,720	2,865	low	29
Gilling	Yorkshire, North Riding	6,030	3,115	low	16
Guisborough	Yorkshire, North Riding	5,720	2,560	med	11
Hadrian's Wall, east end (Wallsend)	Northumberland	5,060	2,850	med	27

Place	County	x	y	Read	km
Hadrian's Wall, west end (Bowness-on-Solway)	Cumberland	5,320	3,740	med	34
Helperby	Yorkshire, North Riding	6,230	2,880	low	8
Kendal	Westmorland	6,080	3,680	low	6
Kirkby Lonsdale	Westmorland	6,210	3,620	low	3
Lancaster	Lancashire	6,380	3,700	med	5
Leeming Bar	Yorkshire, North Riding	6,040	2,995	low	4
Liverpool	Lancashire	7,050	3,820	low	2
Malton	Yorkshire, North Riding	6,430	2,540	low	28
Penrith	Cumberland	5,650	3,640	low	13
Pickering	Yorkshire, North Riding	6,280	2,530	low	26
Pontefract	Yorkshire, West Riding	6,560	2,895	low	19
Preston	Yorkshire	6,690	3,680	med	3
Richmond	Yorkshire, North Riding	6,100	3,180	low	20
Scarborough	Yorkshire, North Riding	6,190	2,420	low	30
Settle	Yorkshire, West Riding	6,230	3,440	low	9
Shap	Westmorland	5,880	3,640	low	8
Skipton	Yorkshire, West Riding	6,210	3,330	low	24
Wakefield	Yorkshire, West Riding	6,505	3,030	low	25
Warrington	Lancashire	7,060	3,550	low	2
Whitby	Yorkshire, North Riding	5,940	2,480	med	21
Wigan	Lancashire	6,880	3,580	low	3
York	Yorkshire, West Riding	6,420	2,780	high	16
Bass Rock	Scotland	3,850	3,410	med	49
Isle of May	Scotland	3,550	3,080	med	83
Orkney (Mainland)	Scotland	1,900	2,050	low	246
Perth	Scotland	4,140	3,720	low	68
Abingdon	Berkshire	8,910	2,680	low	16
Alton	Hampshire	9,500	2,070	high	21
Arundel	Sussex	9,704	1,765	low	19
Barkway	Hertfordshire	8,290	1,645	high	16
Battle	Sussex	9,550	1,095	low	14
Boreham Street	Sussex	9,605	1,275	med	5
Bramber	Sussex	9,686	1,622	low	16
Brentford	Middlesex	8,800	1,835	med	23
Burford	Oxfordshire	8,830	2,810	low	8
Canterbury	Kent	9,075	665	high	22
Chichester	Sussex	9,715	1,815	high	29
Colnbrook	Buckinghamshire	9,020	2,020	med	4
Dunstable	Bedfordshire	8,560	2,020	high	4
Faringdon	Berkshire	9,020	2,720	med	11
Farnham	Surrey	9,420	2,010	high	16
Guildford	Surrey	9,300	1,800	low	20
Havant	Hampshire	9,740	1,990	med	25
High Barnet	Hertfordshire	8,790	1,730	med	8
Hungerford	Berkshire	9,380	2,570	low	23
Kingston	Surrey	9,020	1,845	low	9
Lewes	Sussex	9,672	1,550	med	1
London	Middlesex	8,840	1,610	high	16
Maidenhead	Berkshire	9,045	2,150	med	4
New Alresford	Hampshire	9,545	2,120	med	27
Newbury	Berkshire	9,300	2,550	med	12
Oxford	Oxfordshire	8,790	2,490	high	3
Reading	Berkshire	9,060	2,440	med	15
Royston	Hertfordshire	8,320	1,750	high	8
Rye	Sussex	9,375	860	med	26
Southampton	Hampshire	9,780	2,235	high	31
St Albans	Hertfordshire	8,665	1,855	high	8
Stony Stratford	Buckinghamshire	8,470	2,270	med	4
Tetsworth	Oxfordshire	8,810	2,260	med	9

Place	County	x	y	Read	km
Uxbridge	Middlesex	8,910	2,040	med	9
Waltham Holy Cross	Essex	8,705	1,665	high	8
Ware	Hertfordshire	8,510	1,715	high	14
Winchelsea	Sussex	9,555	1,010	med	7
Winchester	Hampshire	9,590	2,320	high	18
Witney	Oxfordshire	8,800	2,700	low	3
Bridport	Dorset	10,120	3,250	low	25
Bristol	Gloucestershire	9,230	3,190	high	22
Chard	Somerset	9,940	3,515	low	14
Crewkerne	Somerset	9,910	3,285	low	23
Exeter	Devonshire	10,140	3,860	high	18
Gloucester	Gloucestershire	8,890	3,105	high	19
Honiton	Devonshire	10,060	3,690	low	14
Lyme Regis	Dorset	10,170	3,450	low	21
Malmesbury	Wiltshire	9,100	2,950	med	15
Marlborough	Wiltshire	9,300	2,730	med	14
Newent	Gloucestershire	8,740	3,320	med	8
Newport	Gloucestershire	9,100	3,205	med	19
Northleach	Gloucestershire	8,810	2,950	low	7
Okehampton	Devonshire	10,220	4,260	med	19
Salisbury	Wiltshire	9,690	2,600	high	26
Shaftesbury	Dorset	9,760	2,930	med	19
Sherborne	Dorset	9,810	3,225	med	11
Aberdovey	Merionethshire	8,100	5,110	med	66
Abergele	Denbighshire	7,180	4,280	low	10
Aberystwyth	Cardiganshire	8,520	5,100	med	70
Bangor	Carnarvonshire	7,080	4,600	high	24
Barmouth	Merionethshire	7,910	5,100	low	63
Brecon	Brecknockshire	8,880	4,030	med	20
Caernarfon	Carnarvonshire	7,090	4,830	med	38
Capel Curig	Carnarvonshire	7,210	4,510	med	26
Cardigan	Cardiganshire	8,900	4,940	med	33
Carmarthen	Carmarthenshire	9,170	4,640	med	35
Clyro	Radnorshire	8,740	3,830	med	22
Conwy	Carnarvonshire	7,220	4,410	med	5
Criccieth	Carnarvonshire	7,490	5,095	med	52
Flint	Flintshire	7,330	3,960	low	10
Harlech	Merionethshire	7,720	5,130	low	62
Haverfordwest	Pembrokeshire	9,330	5,030	med	43
Llandeilo	Carmarthenshire	8,940	4,510	low	20
Llanenddwyn	Merionethshire	7,800	5,110	low	61
Llangadog	Carmarthenshire	8,950	4,400	low	25
Llawhaden	Pembrokeshire	9,230	4,880	low	36
Llywel	Brecknockshire	8,945	4,240	low	27
Rhuddlan	Flintshire	7,290	4,090	low	8
St Clears	Carmarthenshire	9,200	4,730	low	33
St David's	Pembrokeshire	9,190	5,230	high	37